

Advanced Econometrics – Lab 06 – Mock test

Exercise 1

The inspiration for this exercise was an infographic on Shakespeare's tragedies¹. Your lecturer decided to check how Shakespeare heroes' characteristics affect the way the heroes die. The data set was prepared using the data from the infographics and from an open-source project data².

The dependent variable *death* had 4 categories: 1 - *beheaded*, 2 - *other*, 3 - *poison*, 4 - *stabbed*. The covariates in the model consisted of *written* - year in which the tragedy was written, *speeches* - logarithm of number of speeches of the hero, *words* - logarithm of number of words in the entire play and *gender* - a dummy variable for the hero gender (1 for women and 0 for men).

The output below presents the multinomial logit model results.

Call:

```
mlogit(formula = Death ~ 0 | Written + Speeches + Words, data = dane_mlogit,
        method = "nr")
```

Frequencies of alternatives:

```
beheaded   other   poison   stabbed
0.078125 0.312500 0.062500 0.546875
```

nr method

5 iterations, 0h:0m:0s

$g'(-H)^{-1}g = 2.79E-07$

gradient close to zero

Coefficients :

	Estimate	Std. Error	z-value	Pr(> z)
other:(intercept)	-199.99560297	166.87838699	-1.1985	0.2307
poison:(intercept)	-157.45139170	239.83116281	-0.6565	0.5115
stabbed:(intercept)	-97.76604850	157.80625275	-0.6195	0.5356
other:Written	0.12677157	0.10484938	1.2091	0.2266
poison:Written	0.09626120	0.15036678	0.6402	0.5221
stabbed:Written	0.06390958	0.09924115	0.6440	0.5196
other:Speeches	0.00524247	0.00887017	0.5910	0.5545
poison:Speeches	0.00259439	0.01065403	0.2435	0.8076
stabbed:Speeches	0.00418166	0.00872371	0.4793	0.6317
other:Words	-0.00007453	0.00013218	-0.5638	0.5729
poison:Words	0.00012115	0.00018008	0.6728	0.5011
stabbed:Words	-0.00011628	0.00012779	-0.9099	0.3629

Log-Likelihood: -64.921

McFadden R²: 0.048411

Likelihood ratio test : $\chi^2 = 6.6057$ (p.value = 0.6781)

¹<https://blog.oup.com/2016/04/shakespeare-death-tragedies-infographic/>

²<https://www.opensourceshakespeare.org/>

In all of the following questions apply 10% significance level.

- What was the reason to choose the multinomial logit model instead of the conditional logit?
- Should I have estimated an ordered choice model for the variable *death*?
- What is R^2 equal to? How is it interpreted?
- Are the variables jointly significant? Explain.
- Below you can find combining alternatives tests. Your lecturer decided to combine *stabbed* with *beheaded* into one new category *stabbed&beheaded*. Was it correct? Explain.
- What other categories could be combined besides *stabbed* with *beheaded*? Explain.

**** LR tests for combining alternatives (N=58)

Ho: All coefficients except intercepts associated with a given pair of alternatives are 0 (i.e., alternatives can be collapsed).

Alternatives tested	chi2	df	P>chi2
-----+-----			
beheaded- other	8.187	4	0.085
beheaded- poison	8.333	4	0.080
beheaded- stabbed	2.380	4	0.666
other- poison	4.006	5	0.549
other- stabbed	8.242	4	0.083
poison- stabbed	6.704	4	0.152

The output on the next page presents a multinomial logit model for the variable *death* with two categories combined (*stabbed* with *beheaded*). The questions below relate to the model below.

- The more the hero says, the different her/his death is. Is it true?
- Did Shakespeare take his female heroes' lives in a different way than males' characters? Explain.
- Your lecturer wondered whether Shakespeare's way of taking his heroes' lives evolved in time. What does the output suggest?
- A different hypothesis says that the longer the play, the more unusual ways of killing heroes off in order not to bore readers/spectators. How to verify this hypothesis?

Call:

```
mlogit(formula = Death ~ 0 | Written + Speeches + Words + Gender,
       data = dane_mlogit, reflevel = "stabbed", method = "nr")
```

Frequencies of alternatives:

```
stabbed  other  poison
 0.6250  0.3125  0.0625
```

nr method

6 iterations, 0h:0m:0s

$g'(-H)^{-1}g = 9.5E-07$

gradient close to zero

Coefficients :

	Estimate	Std. Error	z-value	Pr(> z)
other:(intercept)	-108.863139229	93.216387518	-1.1679	0.24287
poison:(intercept)	-45.423578750	205.412453710	-0.2211	0.82499
other:Written	0.066974828	0.058261485	1.1496	0.25033
poison:Written	0.022387028	0.128744243	0.1739	0.86195
other:Speeches	0.003023048	0.003976816	0.7602	0.44715
poison:Speeches	0.002337405	0.007796754	0.2998	0.76434
other:Words	0.000012534	0.000072544	0.1728	0.86282
poison:Words	0.000234582	0.000170395	1.3767	0.16861
other:Gender	1.600752652	0.656331058	2.4389	0.01473 *
poison:Gender	2.935891758	1.327077480	2.2123	0.02695 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -45.516

McFadden R²: 0.14369

Likelihood ratio test : chisq = 15.276 (p.value = 0.054003)